VISION-GUIDED VEHICLE

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OUTLINE

1. Introduction
2. Mechanism Design
3. Mechatronics, code, algorithm
   - Odometry
   - Motor control
   - Encoder calibration
   - Route Algorithm
   - Communication: Raspberry Pi and Arduino
4. Vision
   - Zhang's Camera Calibration
   - Ramer-Douglas–Peucker algorithm
   - Transform Equations
5. Demo
6. Future work

1. INTRODUCTION

- Problem Statement：

  Position Error from **IMU** and **encoder**.

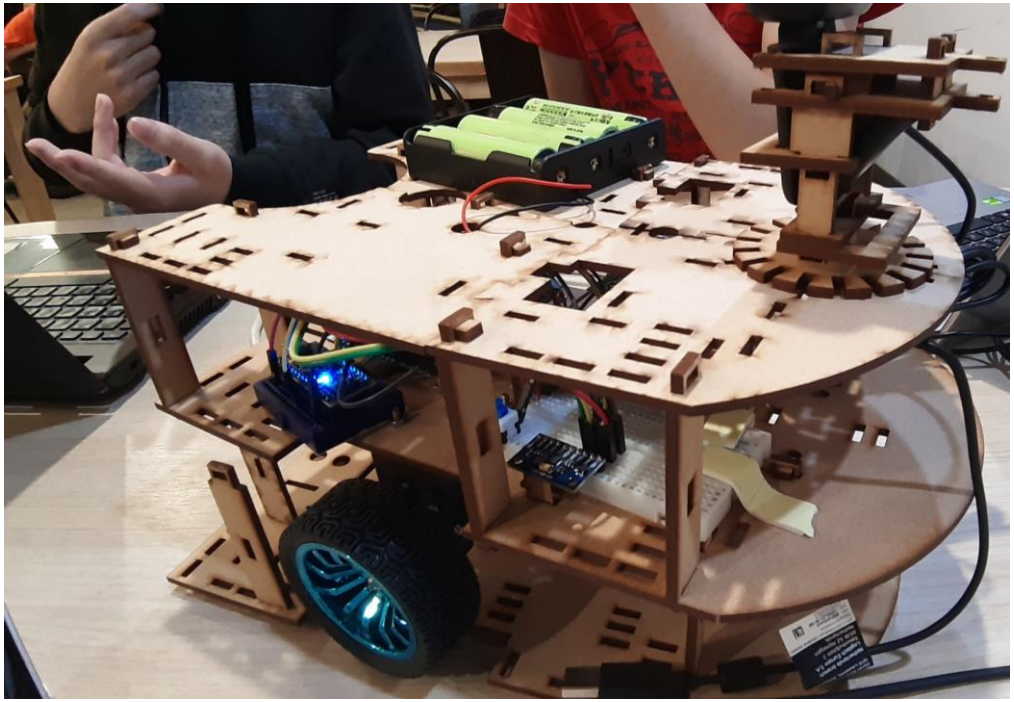
- Solution:

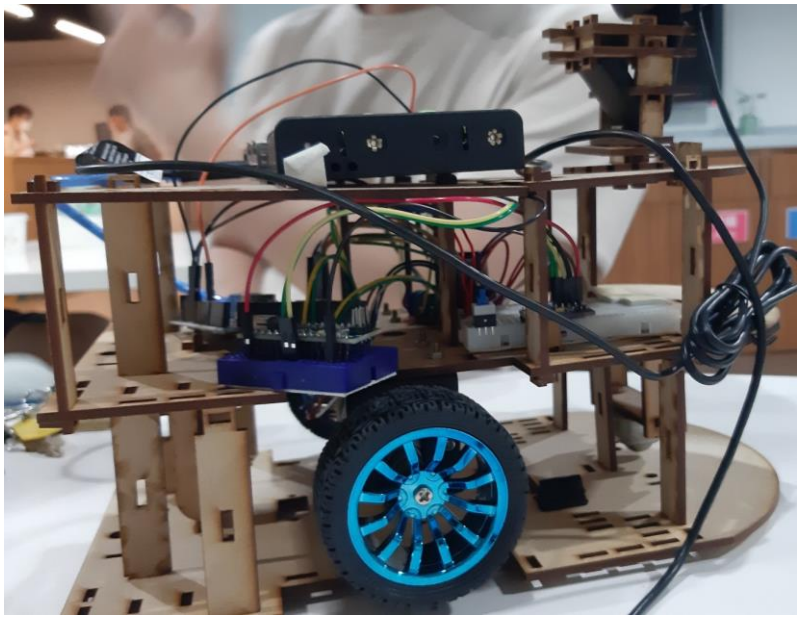
  Apply robot vision **(camara calibration)** to positioning.

- What we made：a vehicle combining a vision-based system
- Vision：to get the world coordinate

2. MECHANISM DESIGN

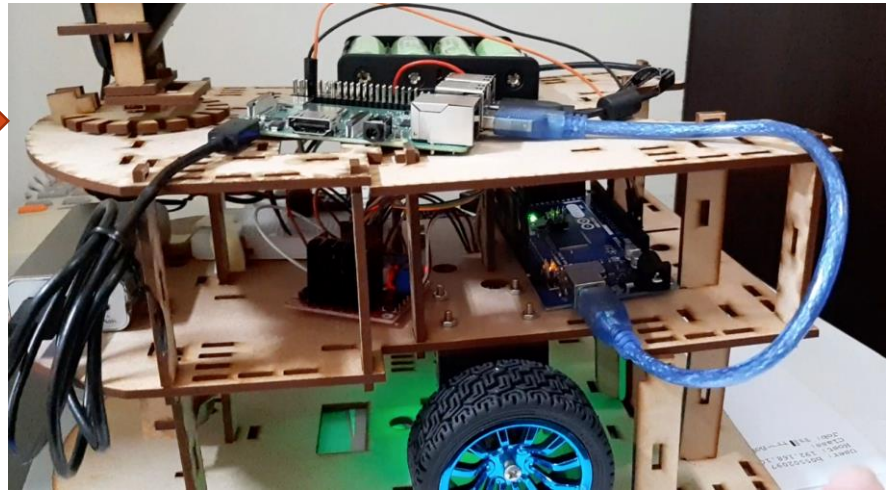
- Overview of the car



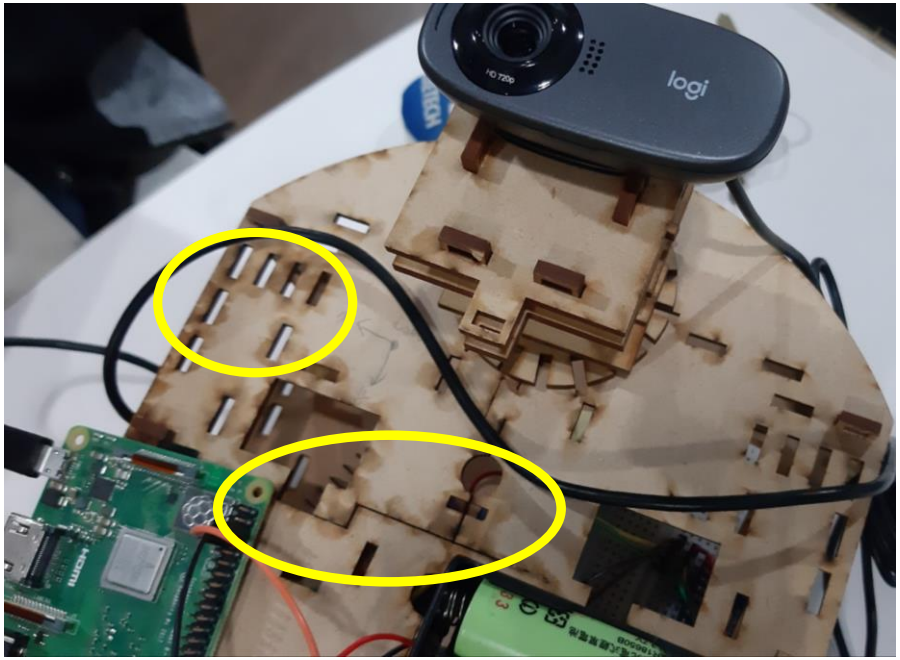


- Designs

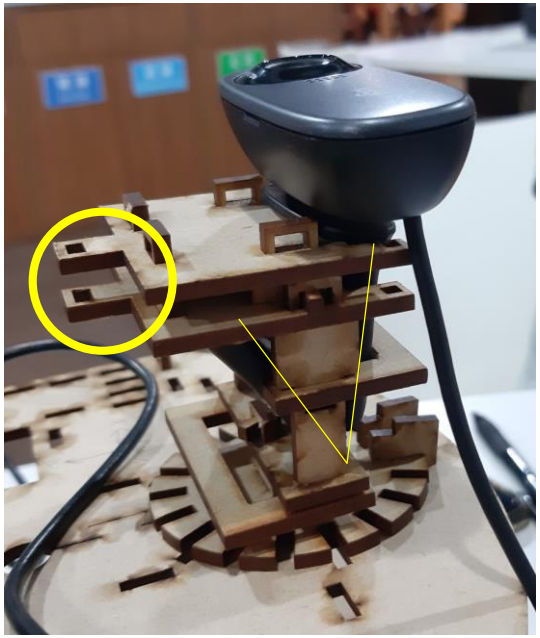
1. Two Layers：

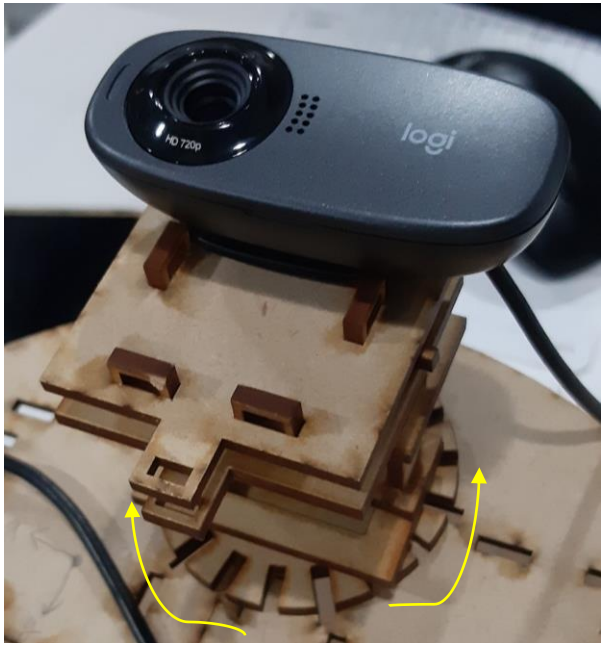


2. Holes & Flexibility：

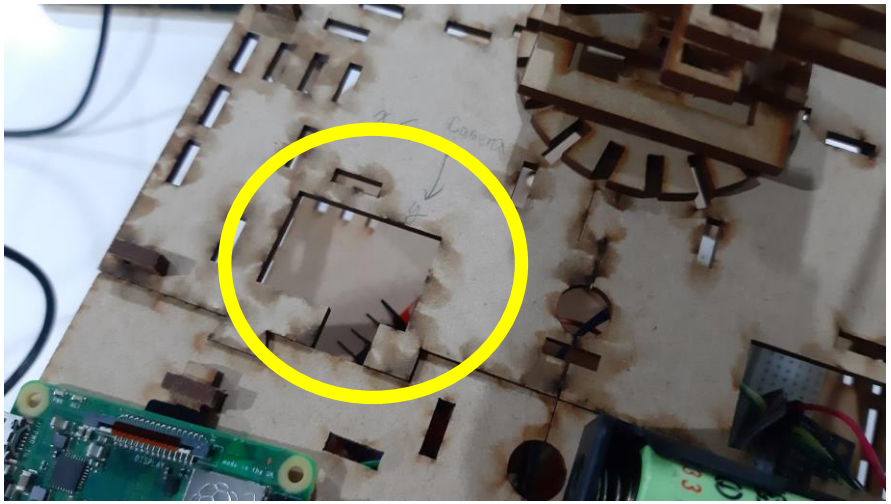


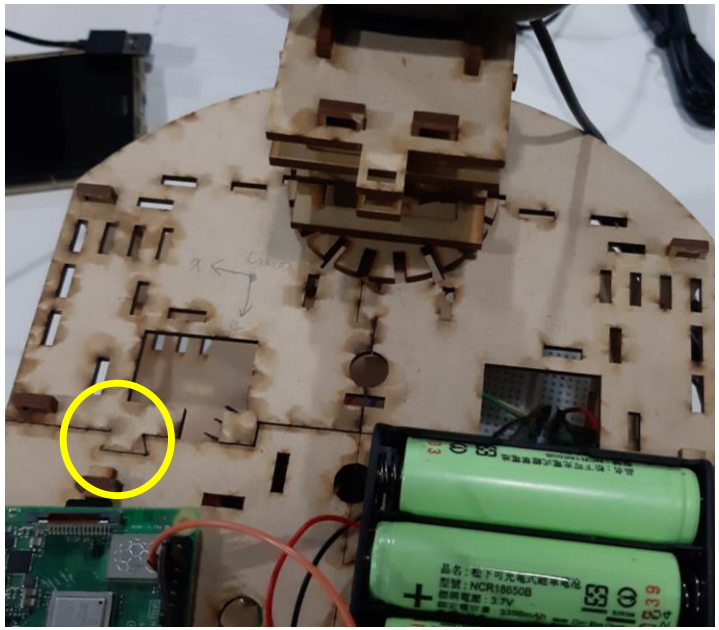
3. Webcam





4. Top Layer



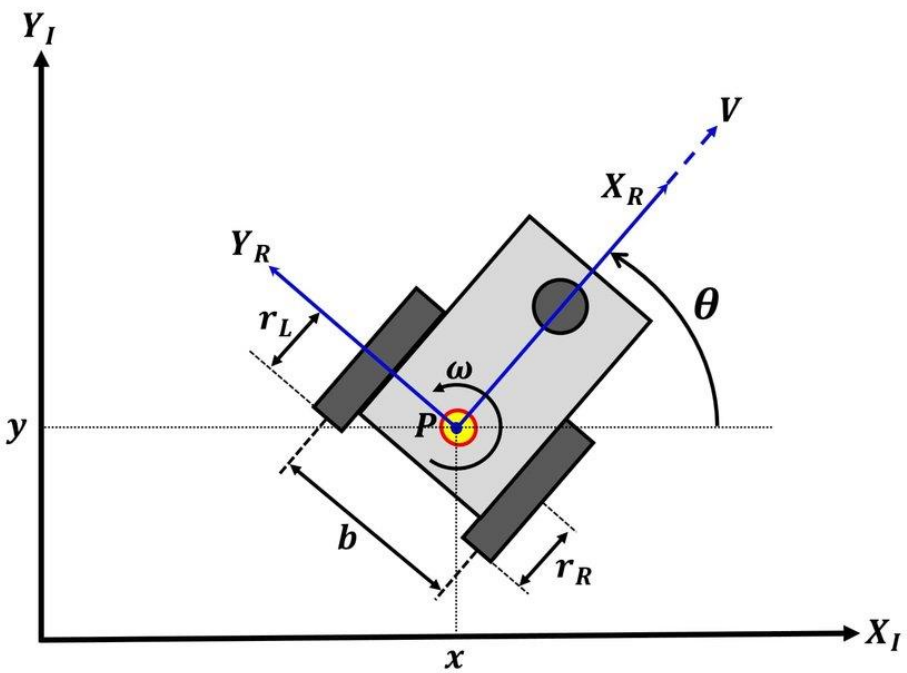


3. MECHATRONICS, CODE, ROUTE ALGORITHM

- Odometry
- Encoder
  - Encoder Calibration
- Route Algorithm
- Communication between Rpi & Arduino
- Motor Control

3-1 ODOMETRY

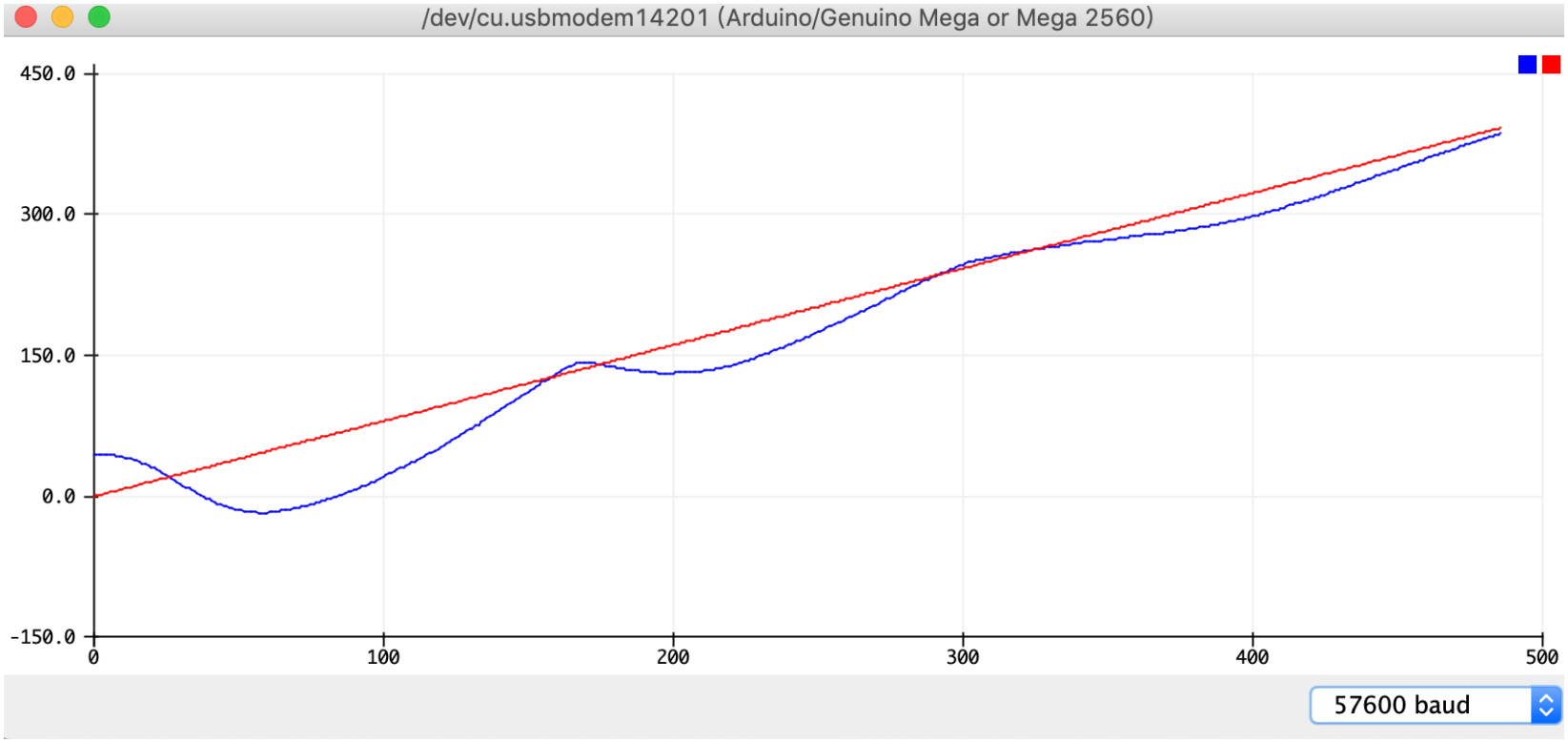
利用左右輪同步轉動達成位移，差速達成車體旋轉，使得車體可以在二為平面上自由移動。



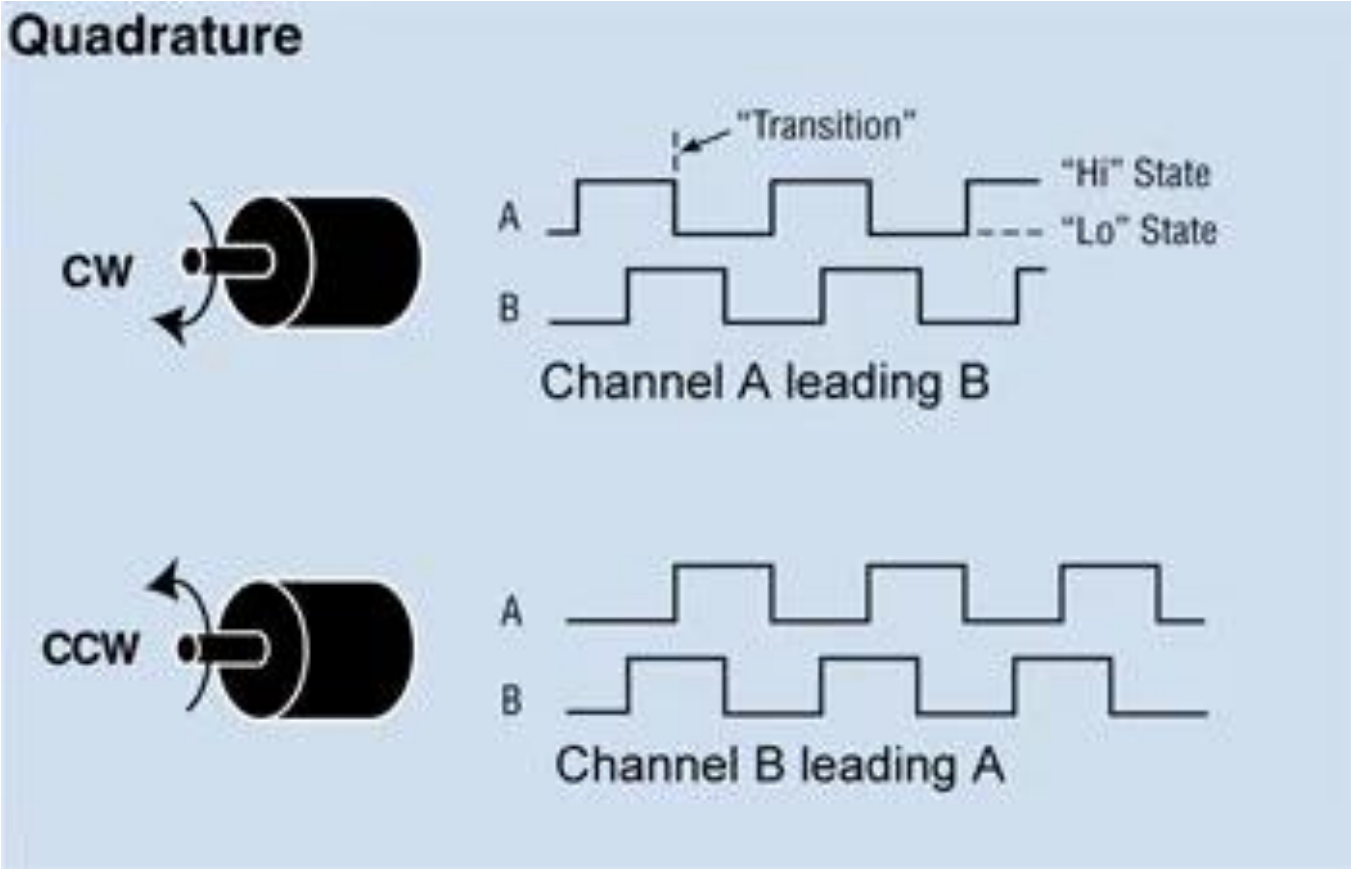
[1]:Nachaya Chindakham

3-2 MOTOR CONTROLL

- PID control with ramp input



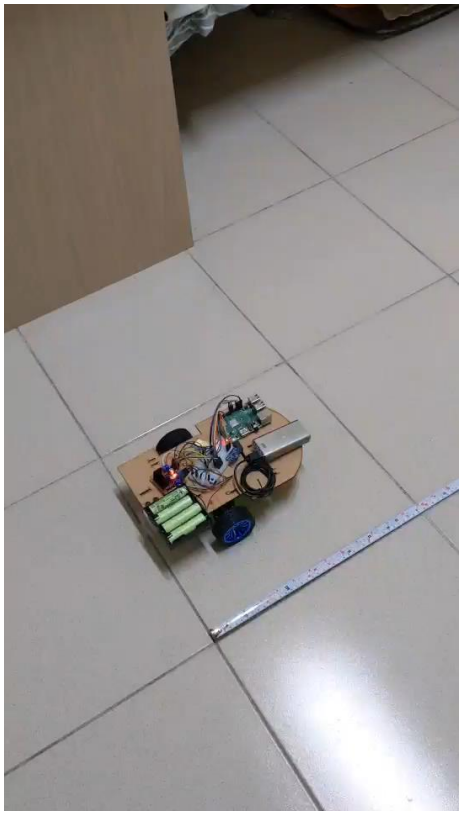
3-3 ENCODER



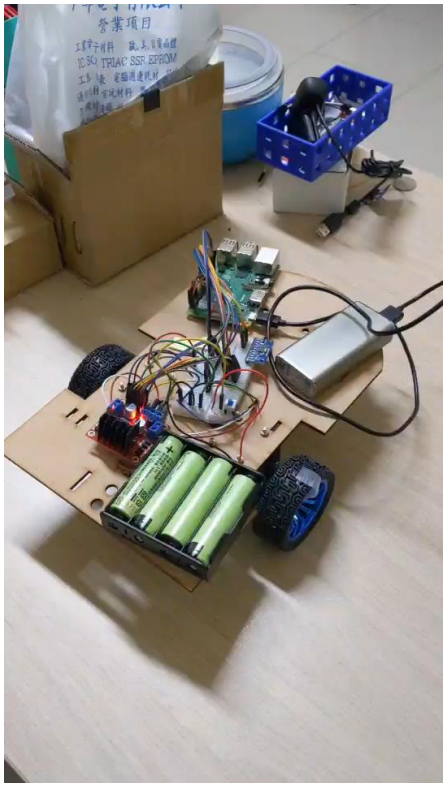
[2] Quadrature Encoder Overview

3-3 ENCODER CALIBRATION

distance

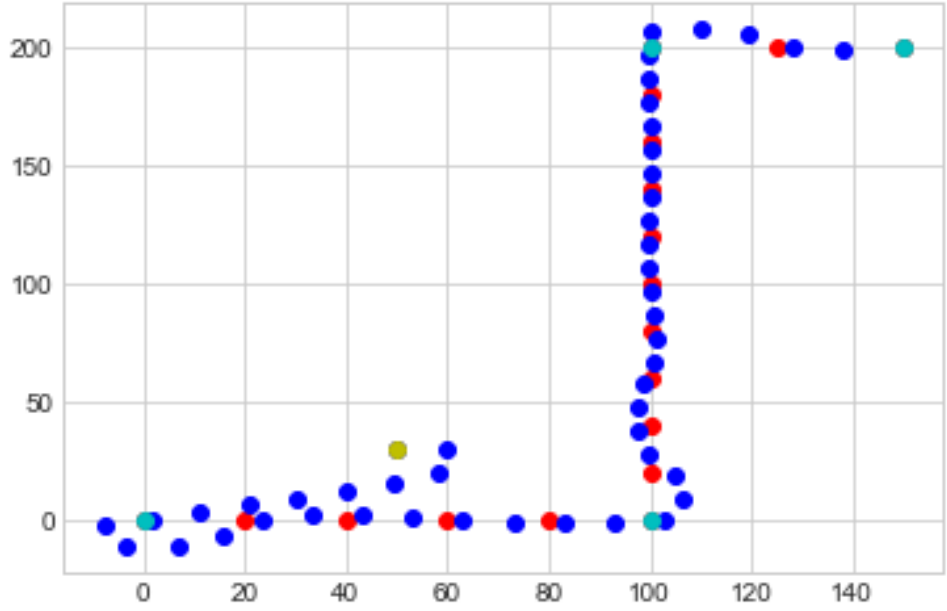


rotation



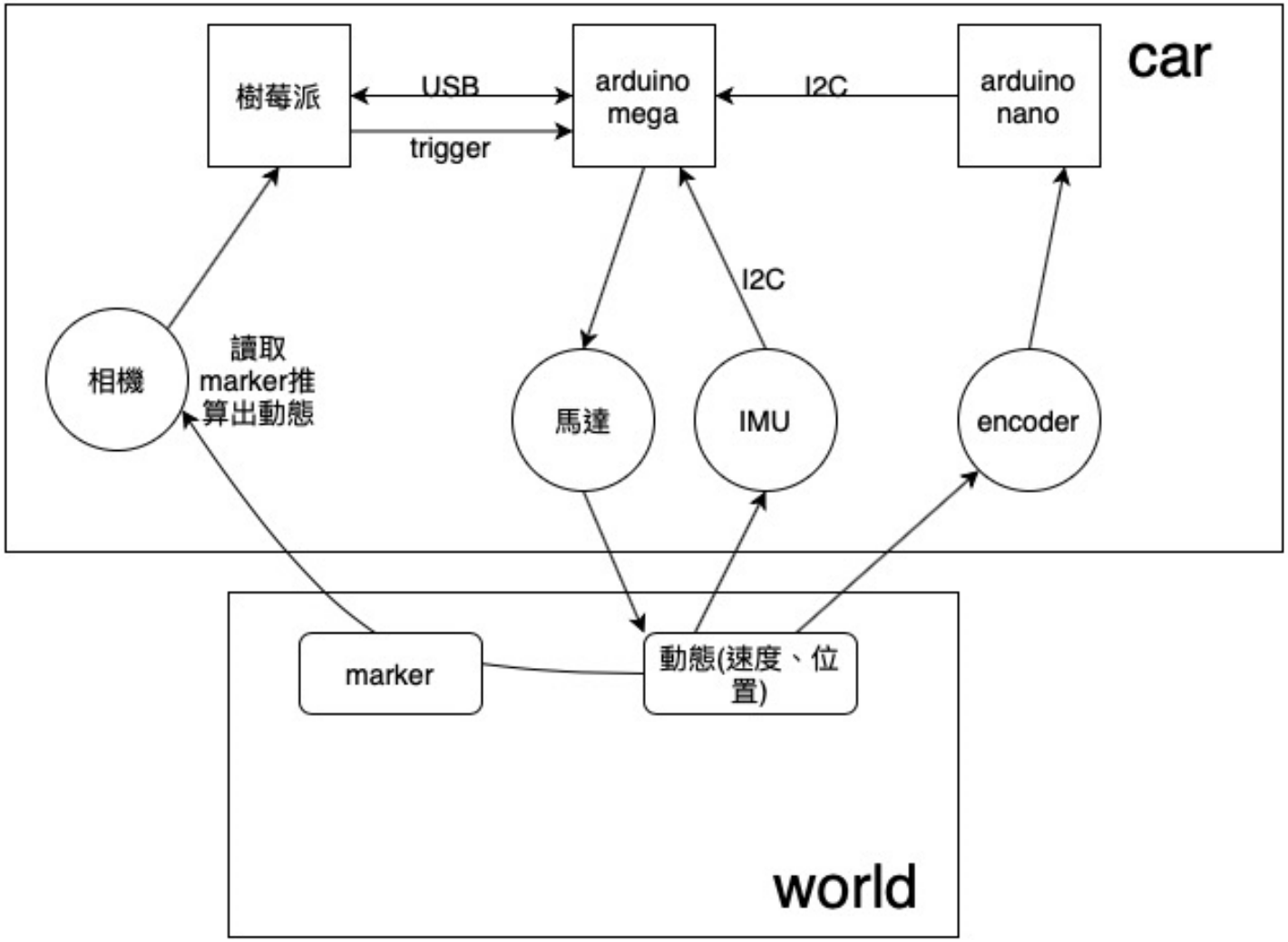
3-4 ROUTE ALGORITHM

- Constant speed
- Changeable omega
- Planning with immediate position and yaw



- Current state is important !!!

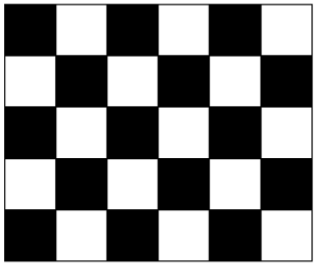
3-5 COMMUNICATION: RASPBERRY PI AND ARDUINO



4. VISION

- Zhang's Camera Calibration
- Ramer-Douglas–Peucker algorithm
- Transform Equations

CAMERA CALIBRATION



- Given the world coordinates in a fixed reference system and the corresponding pixel coordinates of n much larger than 7 feature points
  - **Tsai**'s algorithm establishes an over-determined system of n linear equations based on the radial alignment constraint.
  - **Zhang**'s calibration requires a planar pattern, which defines the lane of a changing world reference system, to be placed at various orientations in front of a camera.

[4][5]

ZHANG'S & TSAI'S

Table 1: Transformations in Tsai and Zhang's models.

Tsai	Zhang
$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \mathbf{R} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + \mathbf{t}$	$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \mathbf{R} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + \mathbf{t}$
$\begin{bmatrix} x_u \\ y_u \end{bmatrix} = \frac{f}{Z_c} \begin{bmatrix} X_c \\ Y_c \end{bmatrix}$	$\begin{bmatrix} x_u \\ y_u \end{bmatrix} = \frac{1}{Z_c} \begin{bmatrix} X_c \\ Y_c \end{bmatrix}$
$\begin{bmatrix} x_u \\ y_u \end{bmatrix} = (1 + k_1^T r^2) \begin{bmatrix} x_d \\ y_d \end{bmatrix}$	$\begin{bmatrix} x_d \\ y_d \end{bmatrix} = (1 + k_1^Z r^2 + k_2^Z r^4) \begin{bmatrix} x_u \\ y_u \end{bmatrix}$
$r = \sqrt{x_d^2 + y_d^2}$	$r = \sqrt{x_u^2 + y_u^2}$
$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = \begin{bmatrix} s_x/d'_x & 0 & c_x \\ 0 & 1/d_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix}$	$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & c_x \\ 0 & \beta & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix}$

- Tsai's Method:
  - Advantage
    - Fast and no initial guess of the camera calibration parameters.
  - Disadvantage
    - Restriction: lens distortion → radial distortion
    - No skewness

- Zhang's Method:
  - At least 3 different projections of a planar calibration target are needed.

[4][5]

ZHANG'S CAMERA CALIBRATION (1/3)

- Zhang's method requires that the calibration object is planar
- The 3D-2D relationship is described by a homography

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{A}\begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \end{bmatrix}\begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix} = \mathbf{A}\begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{bmatrix}\begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} .$$

- Two constraints on the intrinsic parameters

$$\mathbf{h}_1^T \mathbf{A}^{-T} \mathbf{A}^{-1} \mathbf{h}_2 = 0 \tag{3}$$

$$\mathbf{h}_1^T \mathbf{A}^{-T} \mathbf{A}^{-1} \mathbf{h}_1 = \mathbf{h}_2^T \mathbf{A}^{-T} \mathbf{A}^{-1} \mathbf{h}_2 . \tag{4}$$

[4]

ZHANG'S CAMERA CALIBRATION (2/3)

- $B = A^{-T}A^{-1} \equiv \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix}$

  (symmetric)

$$= \begin{bmatrix} \frac{1}{\alpha^2} & -\frac{\gamma}{\alpha^2\beta} & \frac{v_0\gamma - u_0\beta}{\alpha^2\beta} \\ -\frac{\gamma}{\alpha^2\beta} & \frac{\gamma^2}{\alpha^2\beta^2} + \frac{1}{\beta^2} & -\frac{\gamma(v_0\gamma - u_0\beta)}{\alpha^2\beta^2} - \frac{v_0}{\beta^2} \\ \frac{v_0\gamma - u_0\beta}{\alpha^2\beta} & -\frac{\gamma(v_0\gamma - u_0\beta)}{\alpha^2\beta^2} - \frac{v_0}{\beta^2} & \frac{(v_0\gamma - u_0\beta)^2}{\alpha^2\beta^2} + \frac{v_0^2}{\beta^2} + 1 \end{bmatrix}.$$

- $\begin{bmatrix} \boldsymbol{v}_{12}^T \\ \boldsymbol{v}_{11}^T - \boldsymbol{v}_{22}^T \end{bmatrix} \boldsymbol{b} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ where $\mathbf{b} = [B_{11}, B_{12}, B_{22}, B_{13}, B_{23}, B_{33}]^T$.

[4]

ZHANG'S CAMERA CALIBRATION (3/3)

- Given N images of the planar calibration object we stack the equations to get a homogeneous system of linear equations which can be solved by **SVD** when $N \geq 3$
- From the estimated $\boldsymbol{b}$ we can recover all the intrinsic parameters
- The **distortion coefficients** are then estimated solving a linear least-squares problem
- Finally all parameters are refined iteratively

[4]

```
intrinsic = [[659.3,      0.,     331.8],
             [0.,       655.7,    269.0],
             [0.,         0.,       1.]]
distortion = [0.0921015, -0.33639816, 0.01776708, 0.00568066, 1.08561026]
```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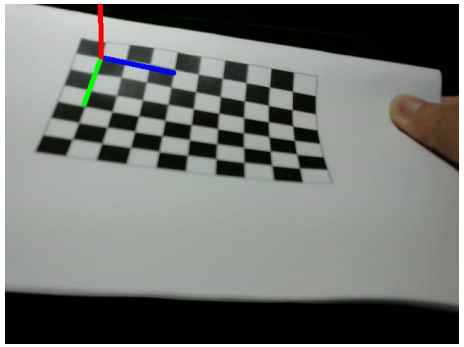
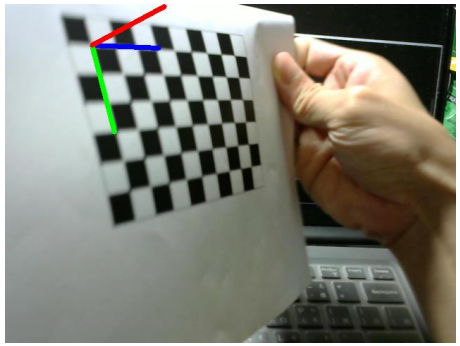
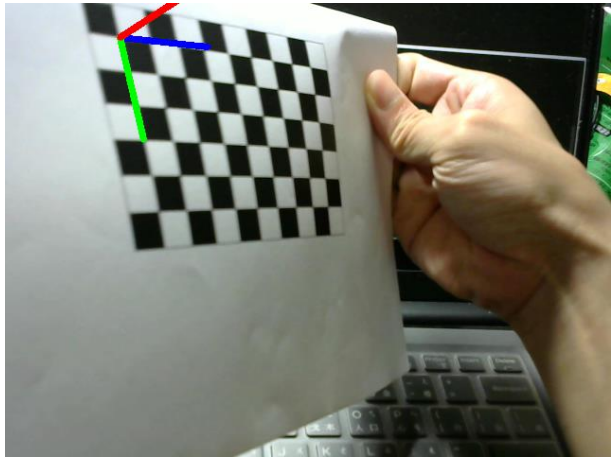
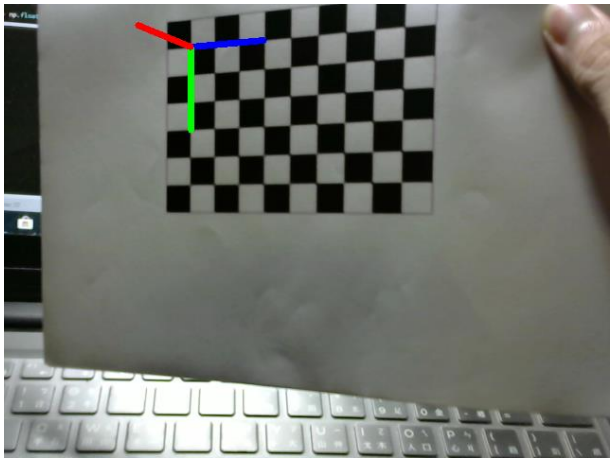
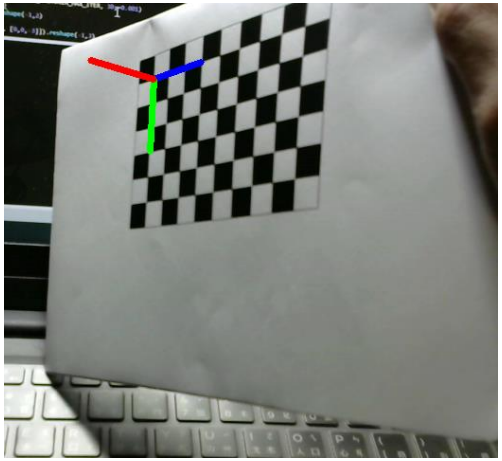
POSE ESTIMATION

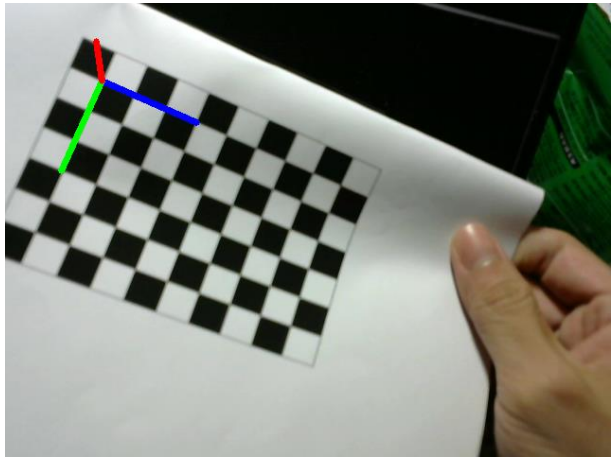


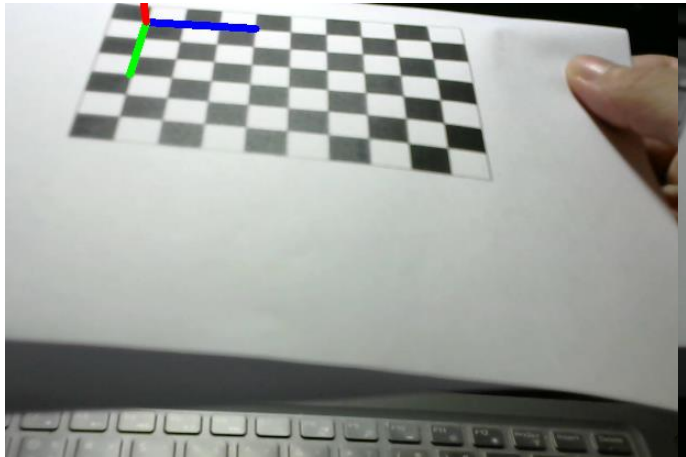


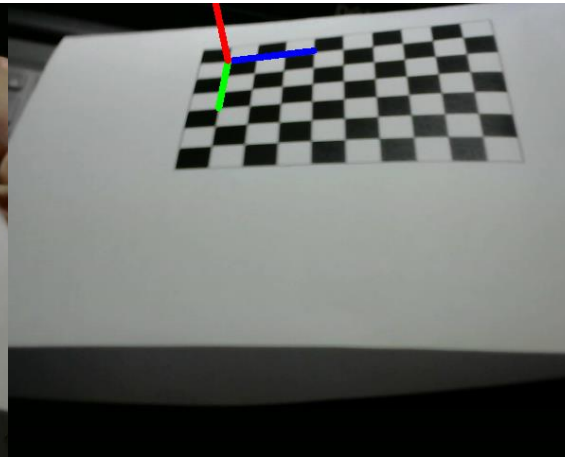






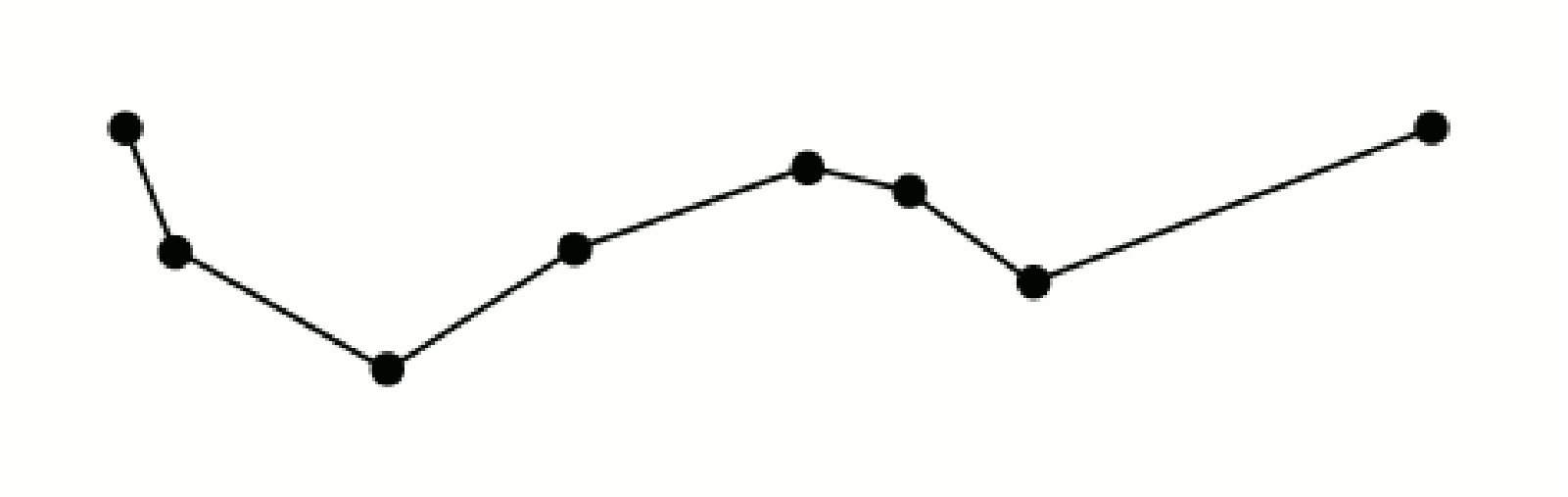






RAMER-DOUGLAS–PEUCKER ALGORITHM

- The Ramer-Douglas–Peucker algorithm is an algorithm for reducing the number of points in a curve that is approximated by a series of points.



[6]

TRANSFORM EQUATIONS

- [3]

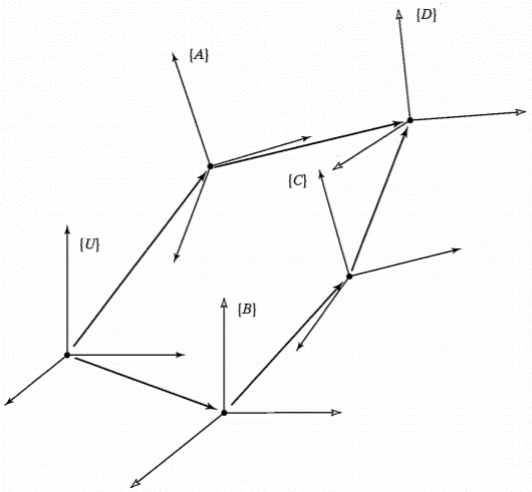


FIGURE 2.14: Set of transforms forming a loop.

Figure 2.14 indicates a situation in which a frame $\{D\}$ can be expressed as products of transformations in two different ways. First,

$$ {}^{U}_{D}T = {}^{U}_{A}T\,{}^{A}_{D}T; \tag{2.48} $$

second;

$$ {}^{U}_{D}T = {}^{U}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T. \tag{2.49} $$

We can set these two descriptions of ${}^{U}_{D}T$ equal to construct a **transform equation**:

$$ {}^{U}_{A}T\,{}^{A}_{D}T = {}^{U}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T. \tag{2.50} $$

INVERTING A TRANSFORM

$$ {}^{B}_{A}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T}\,{}^{A}P_{BORG} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]. \qquad (2.45) $$

$$ {}^{B}_{A}T = {}^{A}_{B}T^{-1}. $$

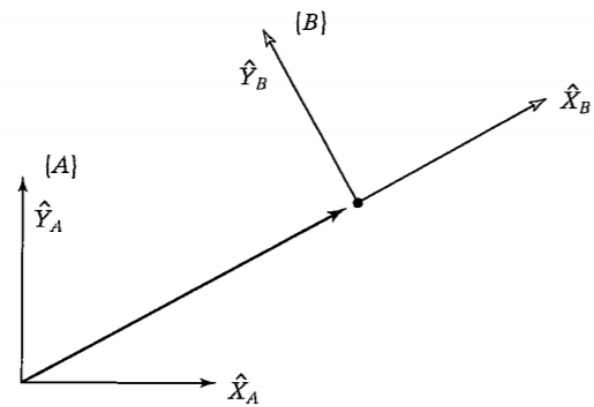
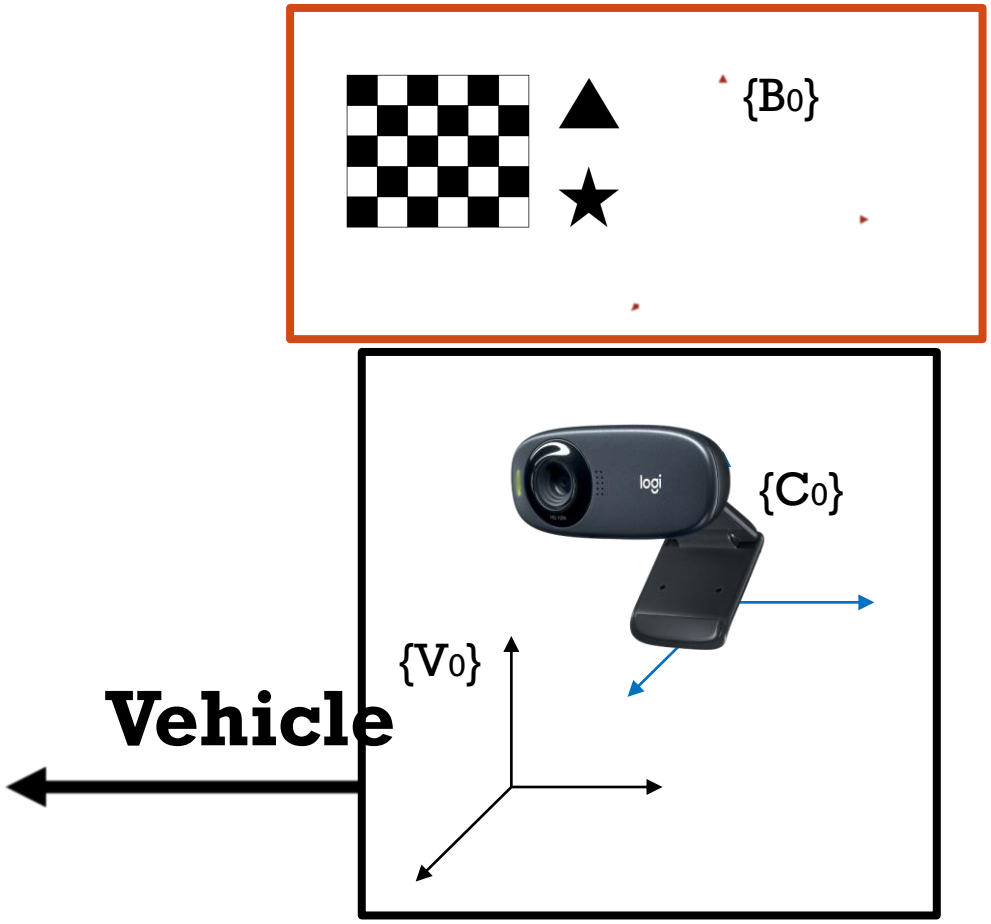


FIGURE 2.13: $\{B\}$ relative to $\{A\}$.

[3]

TRANSFORM EQUATIONS(1/4)

Notation:
0: initial position
J: the jth position
B: Calibration Checker Board
C: Camera
V: Vehicle



https://assets.logitech.com/assets/65486/c310-hd-webcam-

TRANSFORM EQUATIONS(2/4)

Notation:
0: initial position
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B: Calibration Checker Board
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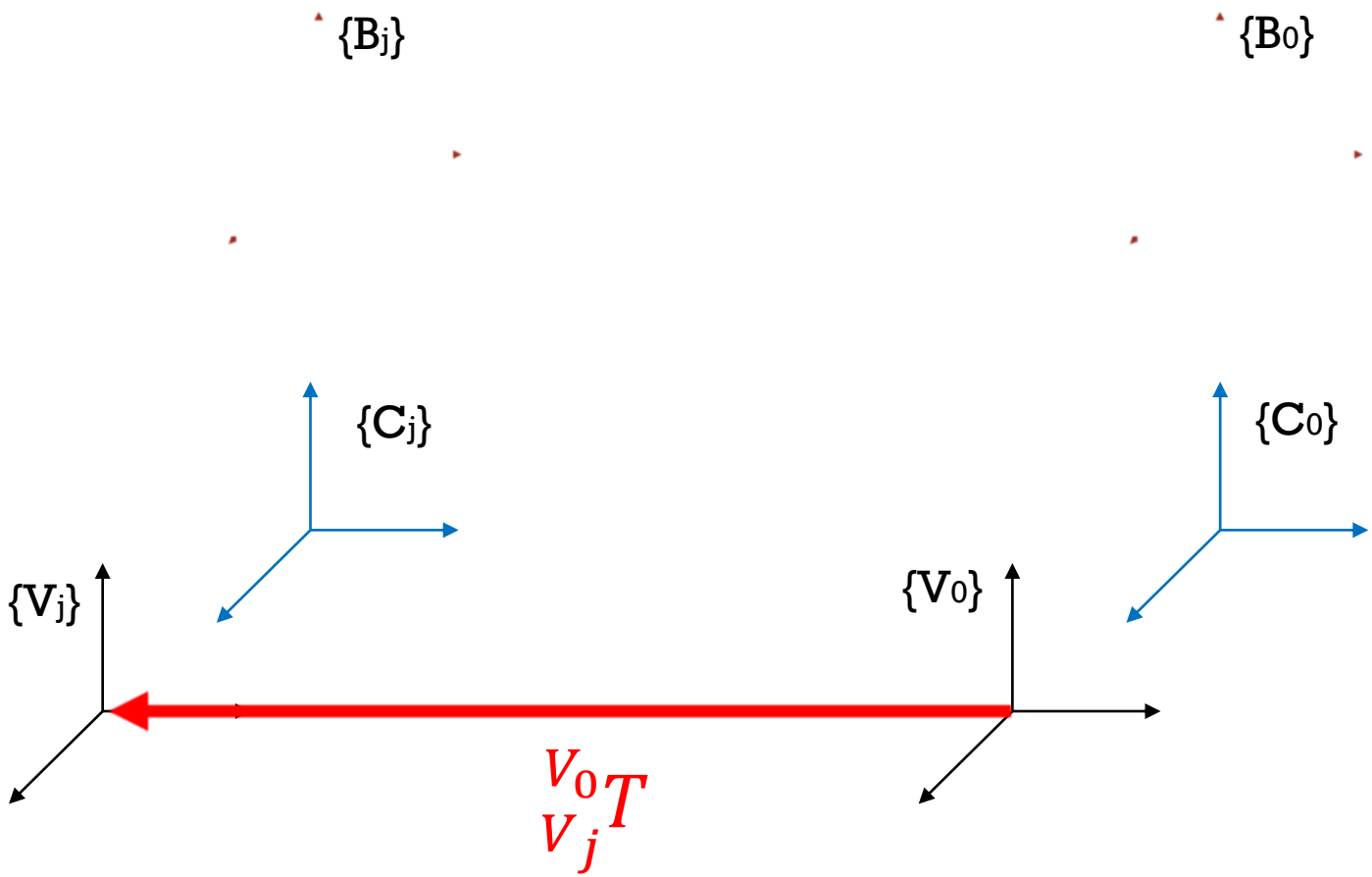
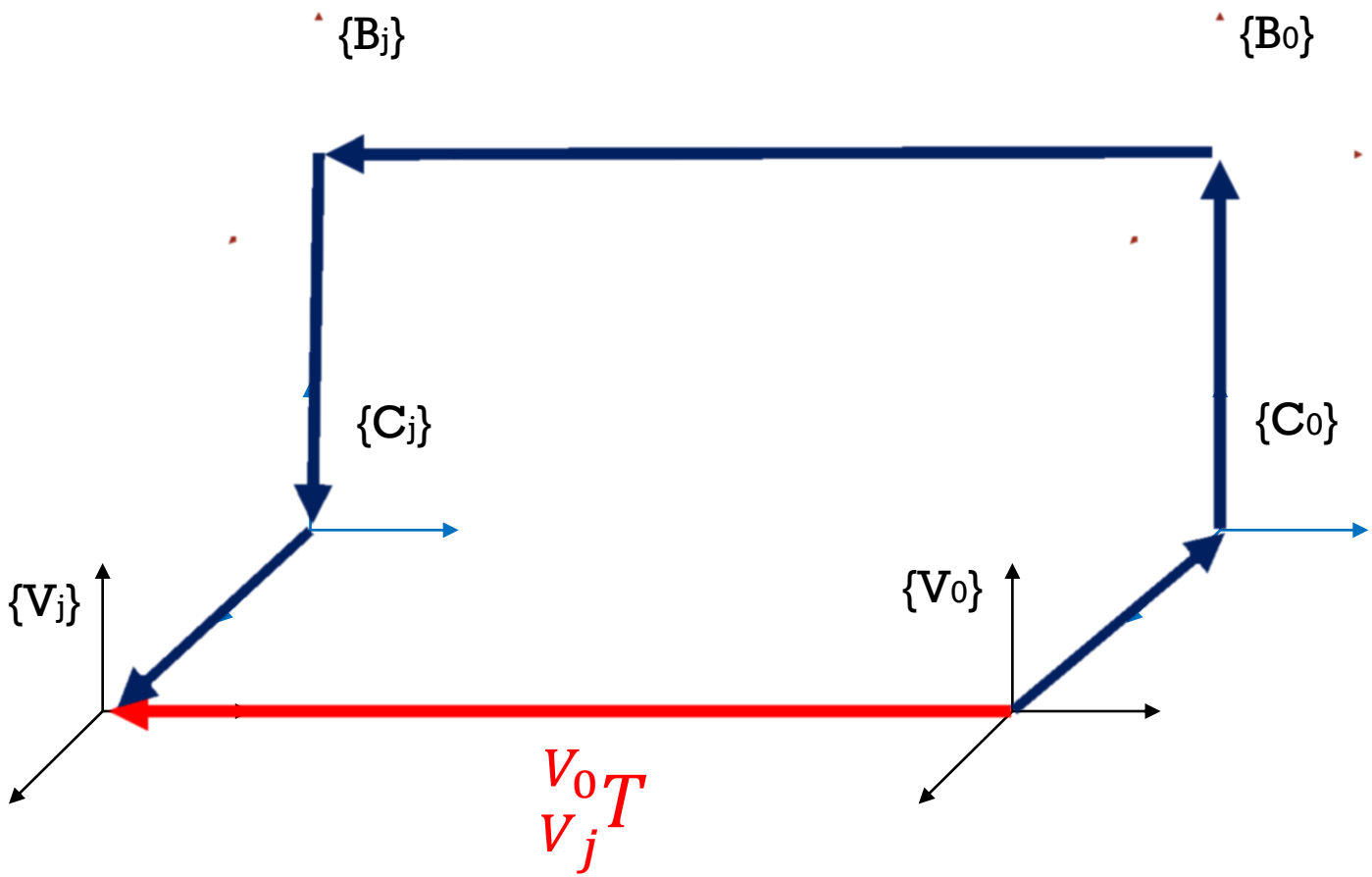


TRANSFORM EQUATIONS(3/4)

Notation:
0: initial position
J: the jth position
B: Calibration Checker Board
C: Camera
V: Vehicle

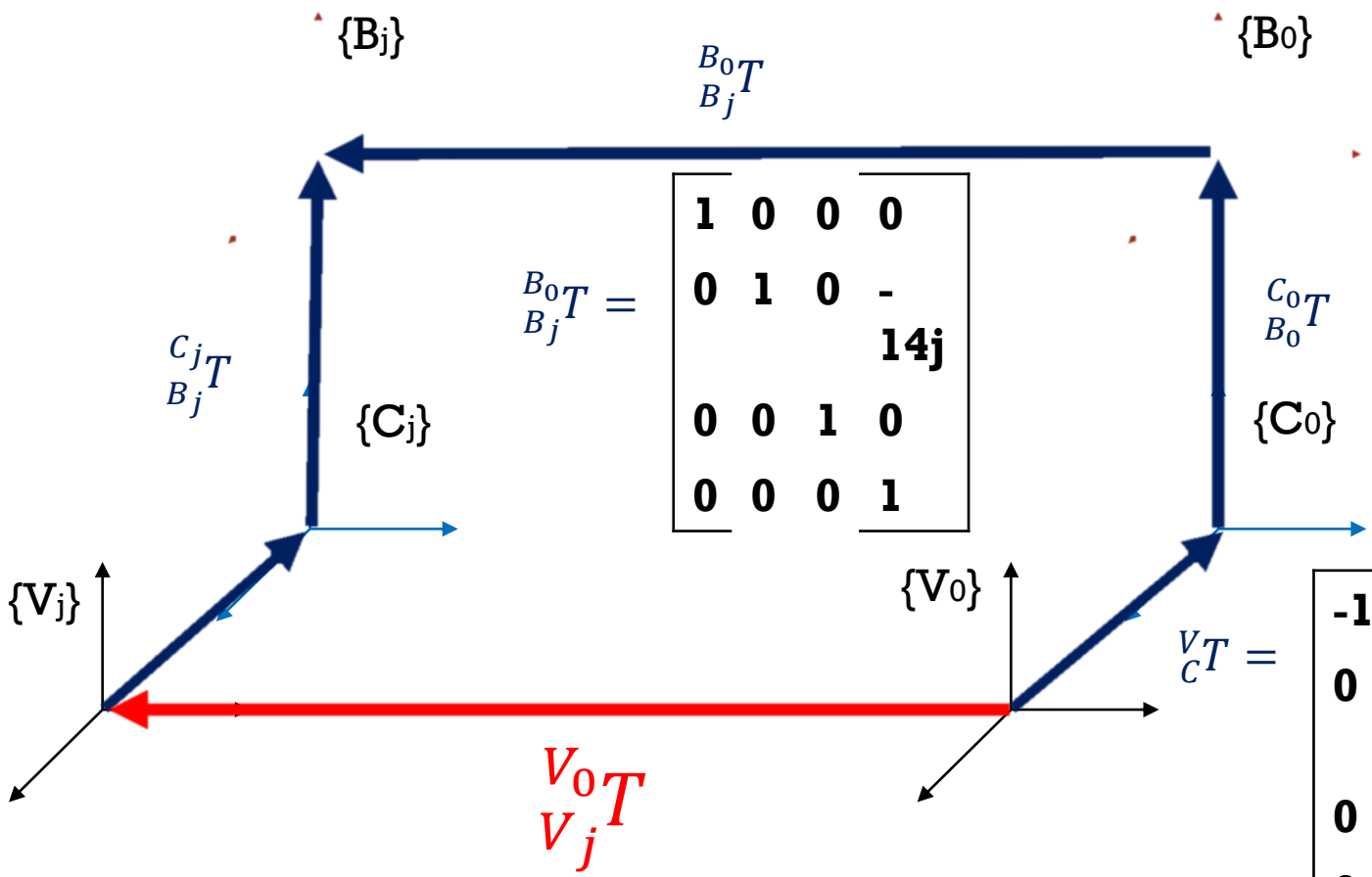
$$ {}^{V_0}_{V_j}T = {}^{V_0}_{C_0}T \; {}^{C_0}_{B_0}T \; {}^{B_0}_{B_1}T \; {}^{B_1}_{C_1}T \; {}^{C_1}_{V_1}T $$



TRANSFORM EQUATIONS(4/4)

Notation:
0: initial position
J: the jth position
B: Calibration Checker Board
C: Camera
V: Vehicle

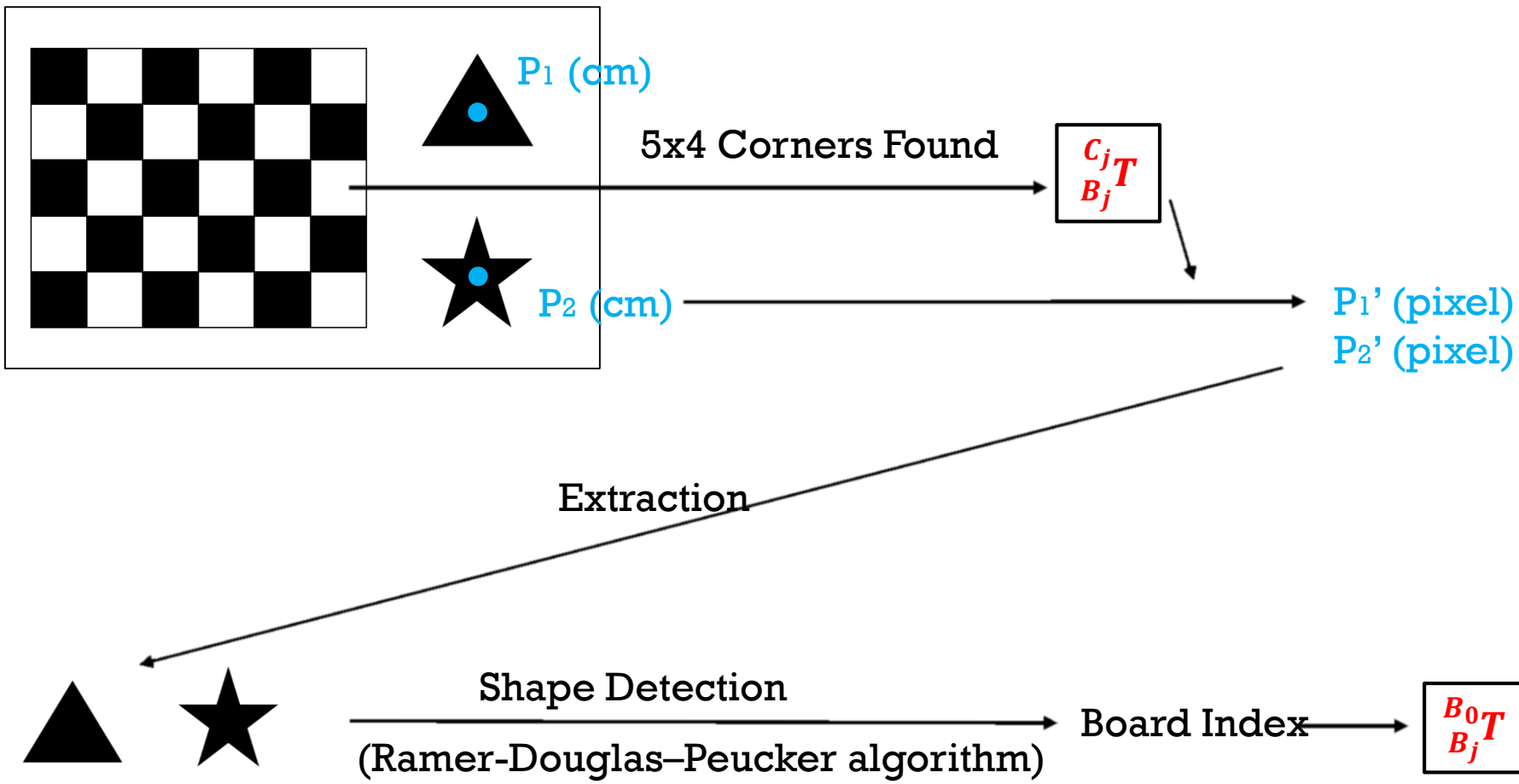
$$ {}^{V_0}_{V_j}T = {}^{V}_{C}T \; {}^{C_0}_{B_0}T \; {}^{B_0}_{B_j}T \left({}^{C_j}_{B_j}T\right)^{-1} \left({}^{V}_{C}T\right)^{-1} $$



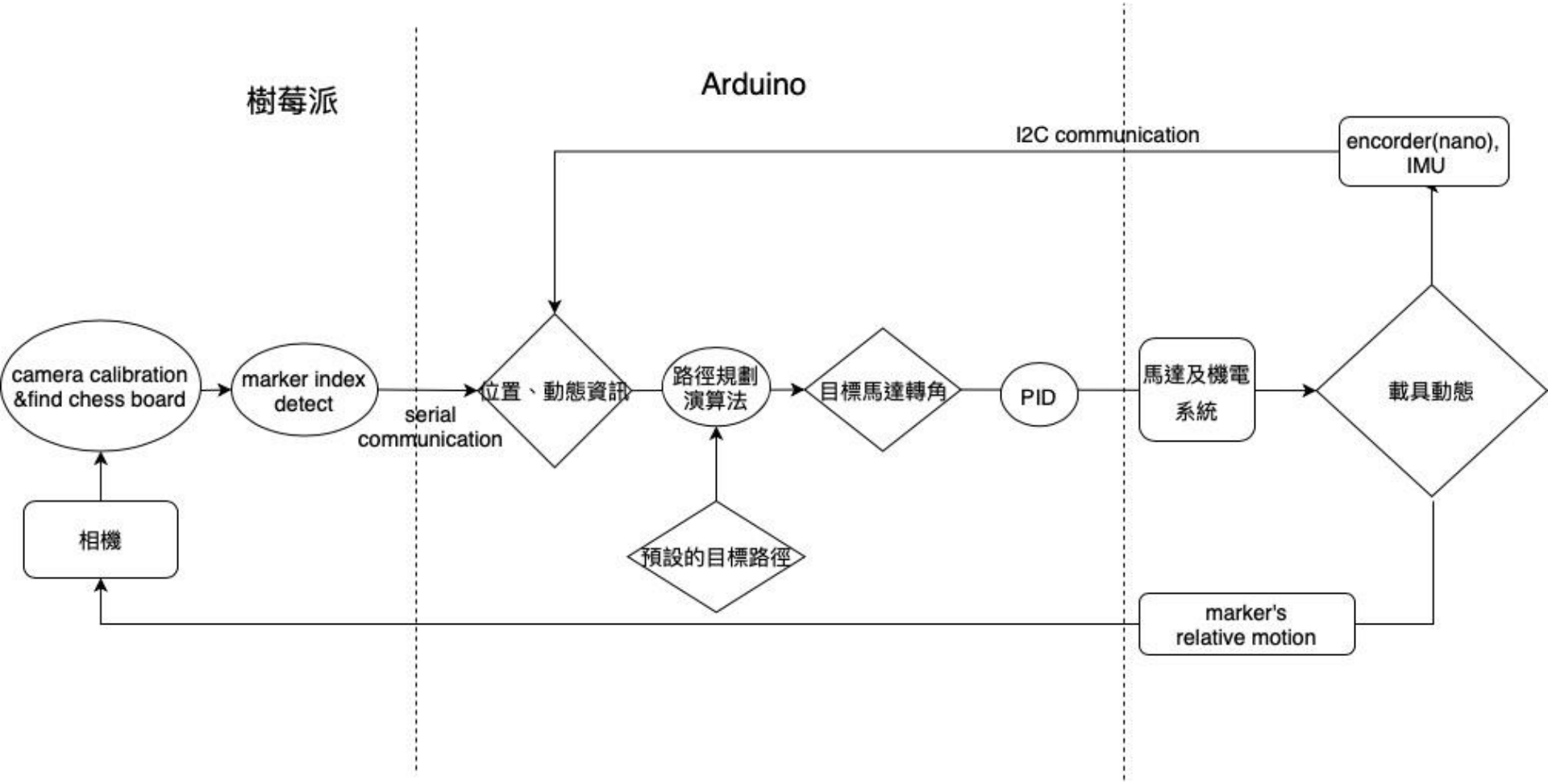
$$ {}^{B_0}_{B_j}T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -14j \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{V}_{C}T = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 12 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{V_0}_{V_j}T = {}^{V}_{C}T \; {}^{C_0}_{B_0}T \; {}^{B_0}_{B_j}T \left({}^{C_j}_{B_j}T \right)^{-1} \left({}^{V}_{C}T \right)^{-1} $$

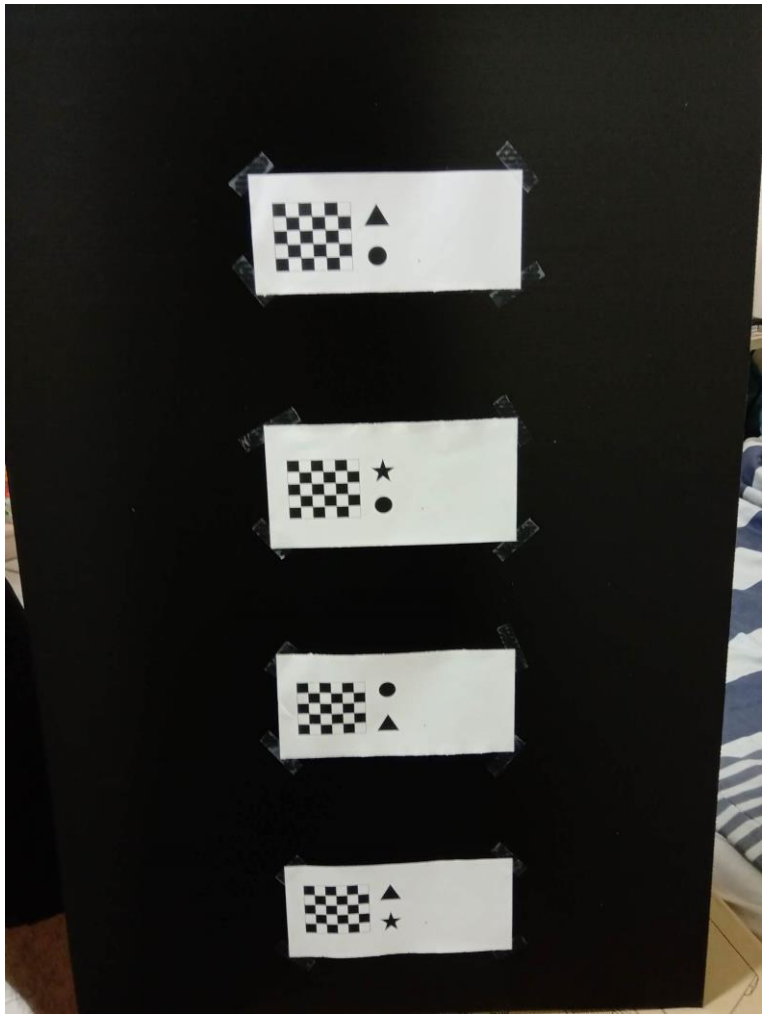


系統整體流程圖

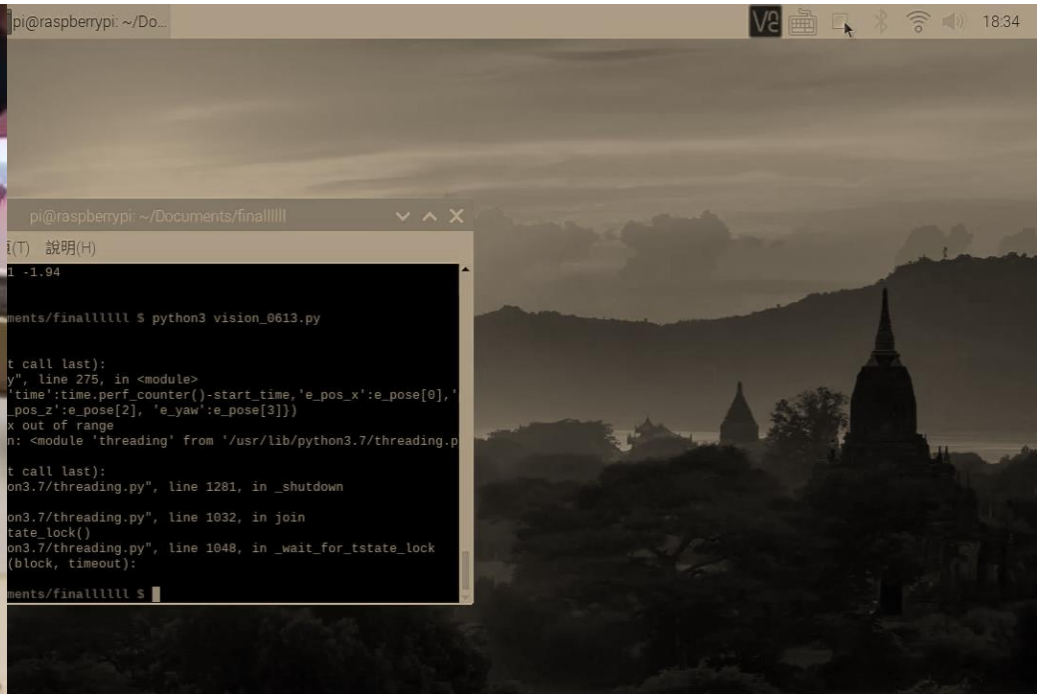

樹莓派
Arduino
I2C communication
encorder(nano), IMU
camera calibration &find chess board
marker index detect
serial communication
位置、動態資訊
路徑規劃演算法
目標馬達轉角
PID
馬達及機電系統
載具動態
相機
預設的目標路徑
marker's relative motion

5. DEMO

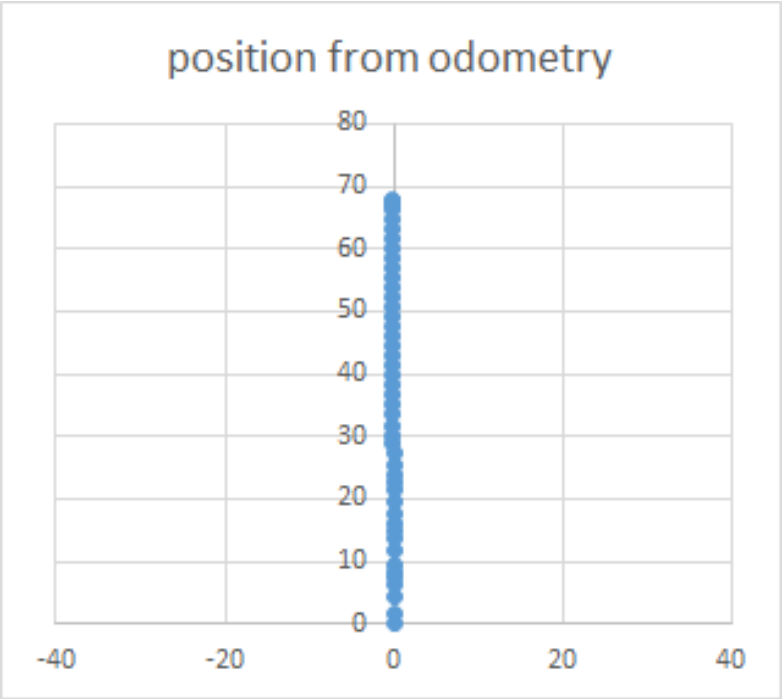
- Target：running 70 cm
- Method：marker board

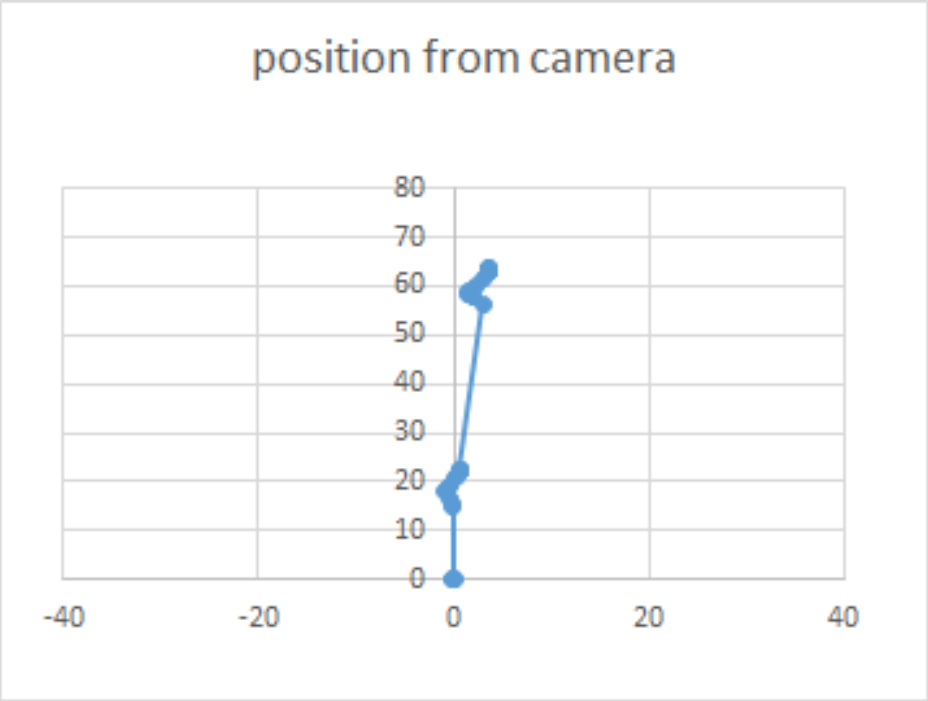


DEMO


pi@raspberrypi: ~/Do...
18:34
pi@raspberrypi: ~/Documents/finallllll
(T) 説明(H)
1 -1.94
ments/finallllll $ python3 vision_0613.py
t call last):
y", line 275, in <module>
'time':time.perf_counter()-start_time,'e_pos_x':e_pose[0],'
_pos_z':e_pose[2], 'e_yaw':e_pose[3]])
x out of range
n: <module 'threading' from '/usr/lib/python3.7/threading.p
t call last):
on3.7/threading.py", line 1281, in _shutdown
on3.7/threading.py", line 1032, in join
tate_lock()
on3.7/threading.py", line 1048, in _wait_for_tstate_lock
(block, timeout):
ments/finallllll $

6. EXPERIMENT RESULT





7. FUTURE WORK

- Real-time feedback from camera calibration.
- Improve the delay caused by the vision algorithm.
- Increase the detection rate of the marker.

分工表

- 王琮文：視覺部分
- 賴橋：路徑規劃演算、馬達控制、通訊、 Raspberry Pi
- 詹澤鋒：機構硬體、報告整合
- 顏貞卿：Arduino、通訊

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- [3] J.J. Craig, *Introduction to Robotics: Mechanics and Control*. Pearson/Prentice-Hall, Upper Saddle River, N.J., 3rd ed edition, 2005.
- [4] Z. Zhang. A flexible new technique for camera calibration. Technical Report MSR-TR-98-71, Microsoft Research, 1998.
- [5] Sun, Wei & Cooperstock, Jeremy. (2005). Requirements for Camera Calibration: Must Accuracy Come with a High Price?. 356-361. 10.1109/ACVMOT.2005.102.
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THANK YOU FOR YOUR ATTENTION